

Timothy Do

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EDUCATION

University of California, Irvine

Bachelor of Science in Electrical Engineering, Accelerated Status

Irvine, CA

Sep. 2019 – June 2023

EXPERIENCE

Undergraduate Student Researcher

University of California, Irvine

January 2022 – Present

Irvine, CA

- Assisted in research under the direction of Professor Glenn Healey on the topic of Spectral Image Filtering for Emissions on Wildfires
- Digitized Imaging Filters into CSVs and compared them in MATLAB against certain chemical emissions.
- Generated altitude maps using QGIS & SRTM 1 Arc-Second Global

Industrial Internet of Things Intern

Western Digital

June 2022 – Sep. 2022

San Jose, CA

- Skills:** MySQL, Python, Raspberry Pi, Soldering, Linux, Lean Six Sigma
- Certified as a Six Sigma Yellow Belt for learning Six Sigma foundations
- Developed Presentation Skills in a Corporate Environment.
- Applied Industrial Tool Sets and Embedded System Programming on Sensor Data Analytics

Vice President

Institute of Printed Circuits Club

Nov. 2021 – Present

Irvine, CA

- Reserved club meeting rooms and sent weekly emails to club members regarding club events
- Programmed the club website and managed social media (i.e. moderator of club discord server)
- Led a Two-Part AM Radio Workshop where the circuit was designed for the best amplifier gain in addition to creating easy-to-read slides & ordering parts from Amazon/Dollar Tree.

PROJECTS

PhotoLabServer | *Python, Matlab*

August 2022

- Implemented a Python Flask Web Server to process images
- Piped with HTTP Post Requests to send images & back with HTTP GET
- Performed image operations like sharpen, smoothing, FFT using MATLAB/Python backend

EECS195-Colorization | *Python, MATLAB*

Feb. 2022 – March 2022

- Colorized grayscale images using GAN and Auto-Encoder Neural Networks
- Gathered own datasets by taking pictures of different scenery
- Clustered pixels into superpixel subgraphs for uniform training weights

TTLZNetChess | *C, Makefile, git*

March 2021 – June 2021

- Managed a team of four developers to develop a C-Based Chess Application
- Implemented Terminal Graphics (Colors) using Escape Codes
- Implemented AIs in Chess using Minimax Algorithm
- Networking Support using standard sockets and FCFS Protocol

PiFmMorse | *Python, bash*

Septemeber 2020

- Programmed a Raspberry Pi to transmit Morse Code with FM over a GPIO Pin
- Concatenated .wav sound files to parse Morse Code messages to transmit
- Called Linux programs to interface with GPIO Pins

TECHNICAL SKILLS

Electronics: Computer Building, Cadence, LTSpice, Soldering, Oscilloscope & Multimeter Measurements

Programming Languages: Java, Python, C, MATLAB, bash, SQL (Postgres), JavaScript, HTML/CSS, VHDL

Developer Tools: Git, SSH, Makefile, Docker, Azure, VS Code, Visual Studio, PyCharm, Eclipse

Libraries: pandas, NumPy, Matplotlib, OpenCV, tensorflow, pytorch